

#Aviation Meteorology

Chapter Seven: Tornadoes and other meteorological Phenomena on aviation



Objectives

- To Be aware of the Effects of Tornadoes on Aviation
- To Know Low And High-level Inversions: Influence On Aircraft Performance
- To Explain the Tropopause influence on aircraft performance

Effects of Tornadoes on Aviation

Airport Infrastructure:

- Tornadoes can cause significant damage to airport infrastructure, including runways, taxiways, hangars, and terminals. This damage may disrupt normal airport operations and lead to the cancellation or diversion of flights.

Flight Disruptions:

- Tornadoes and severe weather associated with them can lead to the suspension or cancellation of flights. Airlines may choose to delay or reroute flights to avoid affected areas, causing disruptions to schedules.

Air Traffic Control (ATC) Challenges:

- Tornadoes and severe weather conditions require careful management by air traffic control. Airports in the affected region may need to reduce or halt operations temporarily, leading to delays and congestion.

Aircraft Safety:

- Tornadoes pose a serious threat to aircraft in flight. While modern aviation technology allows for detection and avoidance of severe weather, the unpredictability and rapid formation of tornadoes can still pose challenges to ensuring the safety of flights.

Preventative Measures:

- Airlines and aviation authorities employ various measures to minimize the impact of tornadoes. This includes monitoring weather conditions, rerouting flights, and implementing safety protocols to protect both aircraft and passengers.

Low And High-level Inversions: Influence On Aircraft Performance

- In aviation, inversions refer to the reversal of the normal lapse rate of temperature with altitude.
- A lapse rate is the rate at which temperature decreases with an increase in altitude. There are two main types of inversions: temperature inversions and temperature lapse rate inversions.
- Let's discuss the influence of both low and high-level inversions on aircraft performance

Low-Level Inversions

- *Temperature Inversions:* At lower altitudes, especially near the surface, temperature inversions can occur. In a standard atmosphere, temperature usually decreases with altitude. However, during a temperature inversion, temperature increases with altitude. This can lead to the trapping of pollutants and moisture near the surface, affecting visibility and air quality.
- *Influence on Aircraft Performance:* Low-level temperature inversions can affect aircraft performance by causing *turbulence, reduced visibility, and decreased engine efficiency*. Pilots may experience a bumpy ride as the inversion layer can create pockets of unstable air.

High-Level Inversions

- High-level inversions generally have a lesser impact on day-to-day aircraft operations compared to low-level inversions. However, they can affect long-range flight planning and fuel efficiency.
- Aircraft flying at high altitudes may experience different atmospheric conditions than those predicted by standard atmospheric models, influencing fuel consumption and range.
- In summary, low and high-level inversions can impact aircraft performance, but their effects are more pronounced at lower altitudes.

Tropopause influence on aircraft performance

- The tropopause is a boundary layer between the troposphere (the lowest layer of the Earth's atmosphere) and the stratosphere.
- Above the tropopause, the stratosphere is characterized by relatively stable and smooth air.
- As a result, aircraft flying in the stratosphere often experience reduced turbulence compared to flights in the troposphere. This can contribute to a smoother ride for passengers and less stress on the airframe

- Aircraft engines are generally more fuel-efficient in the stratosphere due to the stable air conditions. The absence of convective turbulence allows for more consistent engine performance and improved fuel consumption.
- Pilots and flight planners consider the altitude of the tropopause when determining optimal cruising altitudes for long-haul flights. Flying at or near the tropopause can offer advantages in terms of fuel efficiency and reduced weather-related challenges.

#Aviation Meteorology



Chapter Eight: Elements of Forecasting



Objectives

- To Be aware of Preparing the Forecast
- To Identify Types of forecast
- To Know Weather Charts

Preparing the Forecasting

- Forecasting is the process of **predicting** future weather conditions **based** on **current** and **historical** data.
- There are several elements of forecasting that meteorologists and aviation weather specialists consider when making weather forecasts.
- These elements include:-

1.Observations: Observations of current weather conditions are **essential** for making **accurate** weather forecasts.

- ✓ These observations include **surface observations**, such as temperature, dew point, wind speed and direction, and visibility, as well as **upper-level observations**, such as atmospheric pressure and wind speed.

2. Analysis: After collecting observations, meteorologists and aviation weather specialists use **computer** models and other tools to **analyze** the data and identify weather patterns and trends.

- ✓ This analysis can help forecasters **identify** areas of high or low pressure, temperature inversions, and other atmospheric conditions that may **affect** weather conditions.

- **Preparing a weather forecast** involves **analyzing** current and historical weather data and **making** predictions about future weather conditions.
- Different types of weather forecasts are used for **different purposes**, including area forecasts, route forecasts, flight forecasts, local forecasts, and aerodrome forecasts.

- When preparing a weather forecast, meteorologists and aviation weather specialists use a variety of tools and techniques, including computer models, satellite and radar data, and ground-based observations.
- They also consider a range of factors that may affect weather conditions, such as pressure systems, temperature inversions, and local topography.

Types of forecast

1.Area forecast: Area forecasts cover a large geographical area, such as an entire **region** or **country**.

✓ These forecasts are used to provide a **general overview** of weather conditions in a particular area, and it typically cover a **period** of 12 to 24 hours.

2. Route forecast: Route forecasts are used by **pilots** to **plan** their flight paths and **make** decisions about altitude, speed, and fuel consumption.

- ✓ These forecasts cover a **specific flight route** and typically cover a **period** of several hours to a day.

3. Flight forecast: Flight forecasts provide detailed **information** about weather conditions along a **specific flight path**.

- ✓ They include information on cloud cover, turbulence, icing conditions, and other factors that may **affect** the safety and efficiency of the flight.

4. Local forecast: Local forecasts are used to provide **information** about weather conditions in a **particular city** or region.

- ✓ It typically cover a period of 12 to 24 hours and include information on temperature, precipitation, and wind speed and direction.

5. Aerodrome forecast: Aerodrome forecasts provide **information** about weather conditions at a **specific airport** or **airfield**.

- ✓ They include information on visibility, cloud cover, and wind conditions, which are **critical** for safe takeoff and landing operations.

Weather Charts

- Weather charts are **graphical** representations of weather data used by **pilots** and aviation weather specialists to make **informed** decisions about flight operations.
- Different types of weather charts are used to provide **information** about different aspects of weather conditions, including pressure systems, wind patterns, and precipitation.

➤ Some common types of weather charts used in aviation are as following:-

1.Surface analysis charts: Surface analysis charts provide a snapshot of current weather conditions at the Earth's surface.

- ✓ They show areas of high and low pressure, temperature, wind direction and speed, and precipitation.

2. Upper-level charts: Upper-level charts provide information about weather conditions at higher altitudes.

- They show wind speed and direction, temperature, and pressure at different levels of the atmosphere.
- These charts are useful for predicting turbulence and other weather conditions that may affect flight operations.

3. Radar summary charts: Radar summary charts provide information about precipitation patterns over a specific region.

- They show the location and intensity of precipitation and can be used to identify areas of potential turbulence and other hazards.

4. Sigmet charts: Sigmet charts provide information about significant meteorological events, such as thunderstorms, volcanic ash clouds, and severe turbulence.

- They are issued by aviation weather centers and provide critical information for pilots and air traffic controllers.

5. Airmet charts: Airmet charts provide information about weather conditions that may affect the safety and efficiency of flight operations.

- They provide information about low-level turbulence, icing conditions, and other hazards.

7. Significant weather charts: Significant weather charts provide an overview of weather conditions across a broad geographic region, typically covering a range of several hundred miles.

- They show the location and movement of pressure systems, fronts, and other weather features that may affect aviation operations.
- They also highlight areas of significant weather, such as thunderstorms, icing conditions, and turbulence.
- These charts are particularly useful for pilots and air traffic controllers planning long-distance flights.

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SFC SURFACE ANALYSIS

ISOBARS (1000 MB)

FRONT (SOLID LINE WITH TRIANGLES OR SEMICIRCLES)

WAVE (DOTTED LINE WITH SEMICIRCLES)

SHORE (DOTTED LINE WITH SEMICIRCLES)

ICE (DOTTED LINE WITH SEMICIRCLES)

CLIMATE (DOTTED LINE WITH SEMICIRCLES)

1022

1020

1018

1016

1014

1012

1010

1008

1006

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1002

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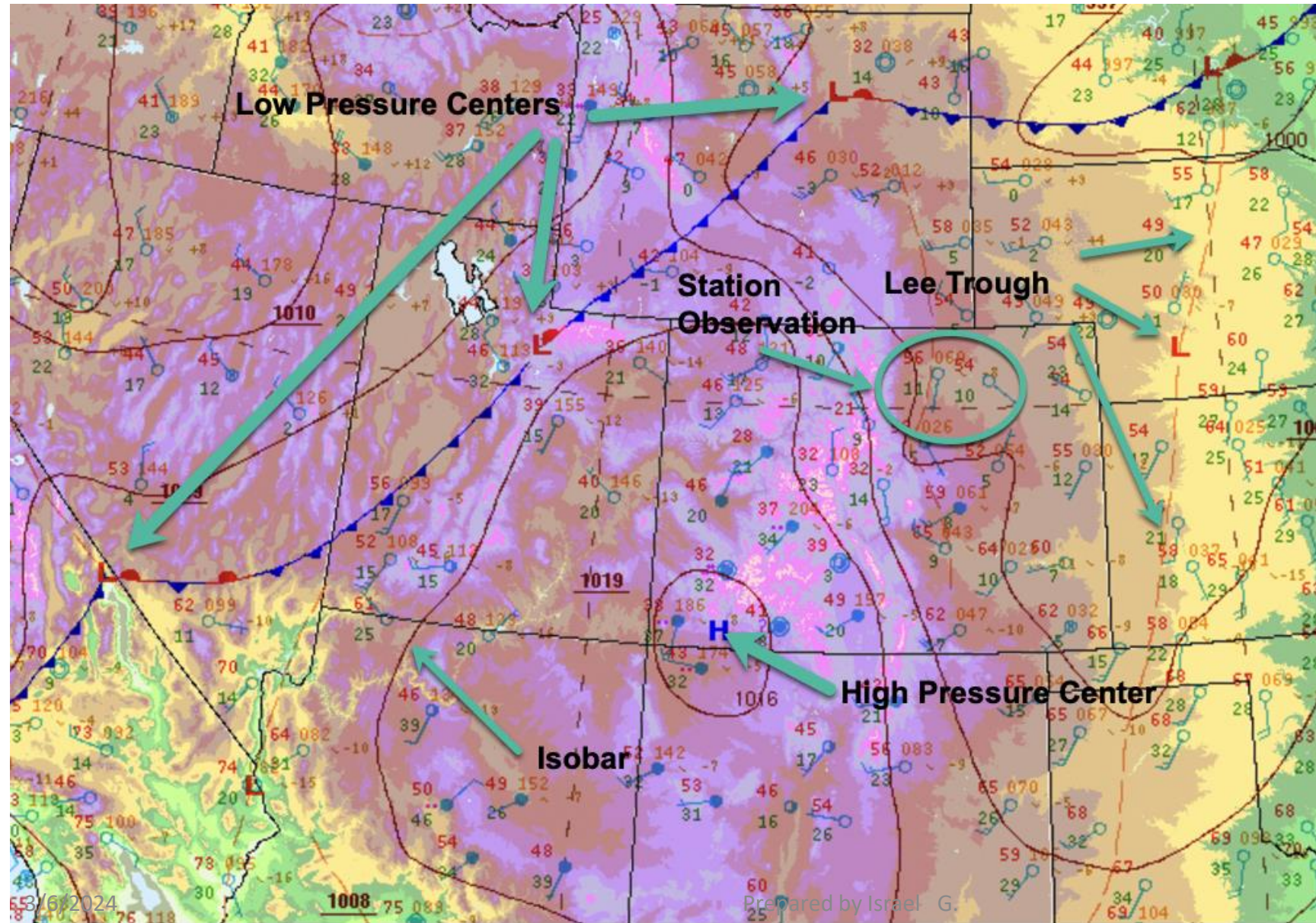
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Figure 5-1. Surface Analysis Chart.

Surface analysis charts

- ✓ Surface analysis charts can give you both a **synoptic**, or large-scale, view of weather features, as well as **mesoscale**, or smaller-scale, features.
- ✓ Surface analysis charts show locations of **fronts, highs and lows, squall lines, dry lines**, and other **weather** phenomena that are important in understanding **current weather** and **predicting** future weather events.

Cont...



Precipitation and Weather Conditions

- Below are some of the symbols used to indicate weather conditions:

			Rain (light, moderate, heavy)
			Snow (light, moderate, heavy)
			Thunder (with rain, snow, no precipitation)
			Shower (rain or snow)
			Drizzle
			Freezing rain, freezing drizzle
			Ice pellets/sleet
			Fog (shallow, deep)
			Haze

Cont...

Pressure tendency symbol and their description



Continuously falling



Falling, then steady



Falling before a lesser rise



Rising before a greater fall



Continuously rising



Rising then steady



Falling before a greater rise



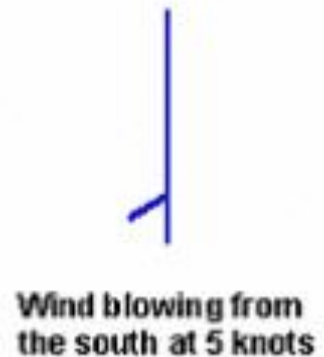
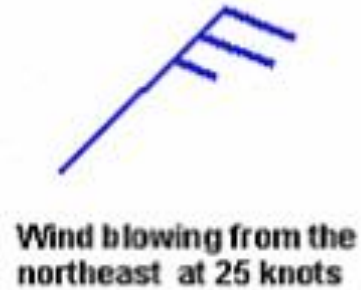
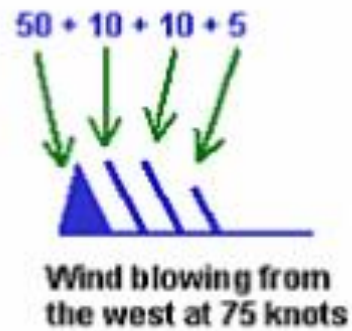
Rising before a lesser fall



Steady

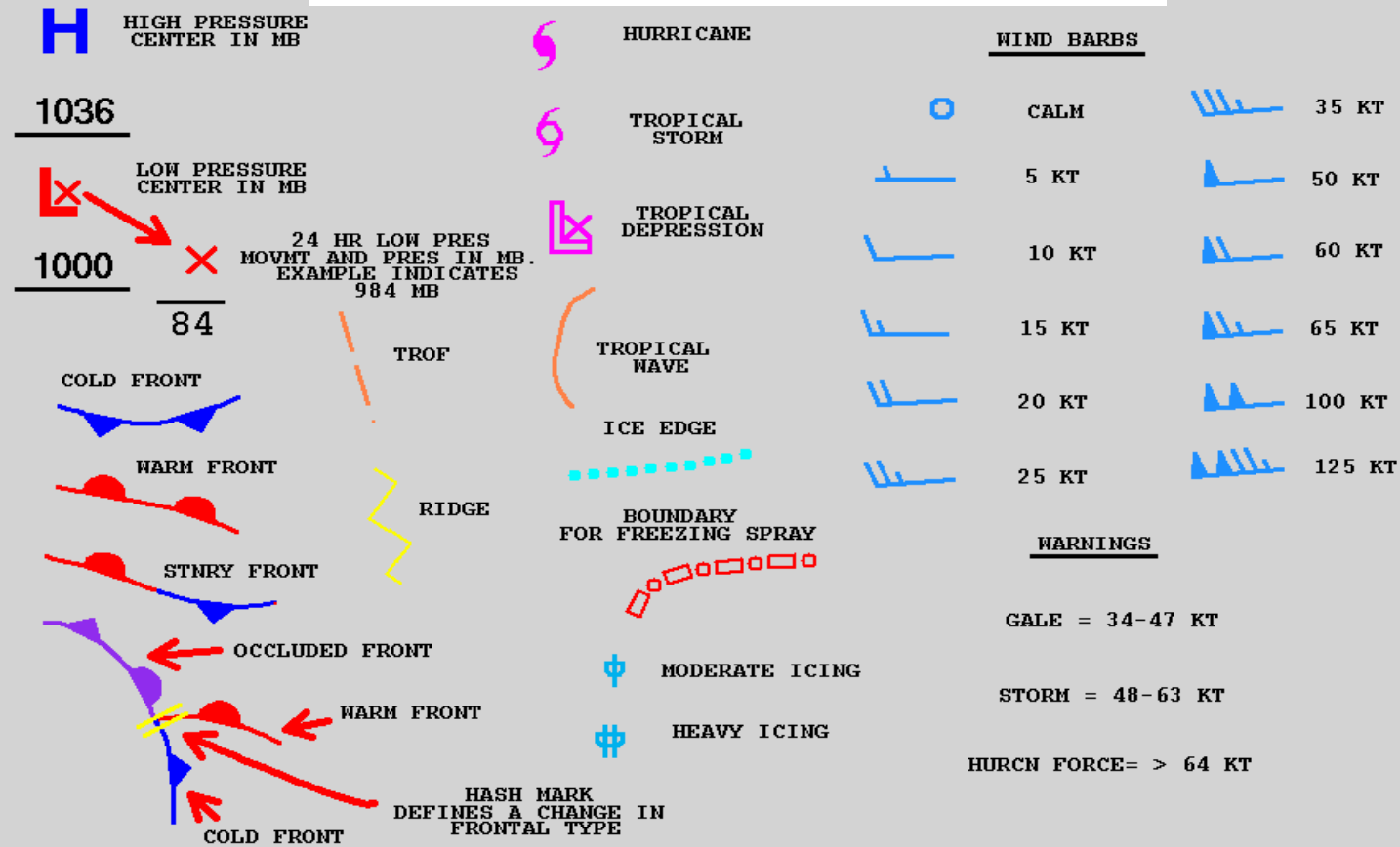
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Wind speed and direction and sky cover symbol and their description

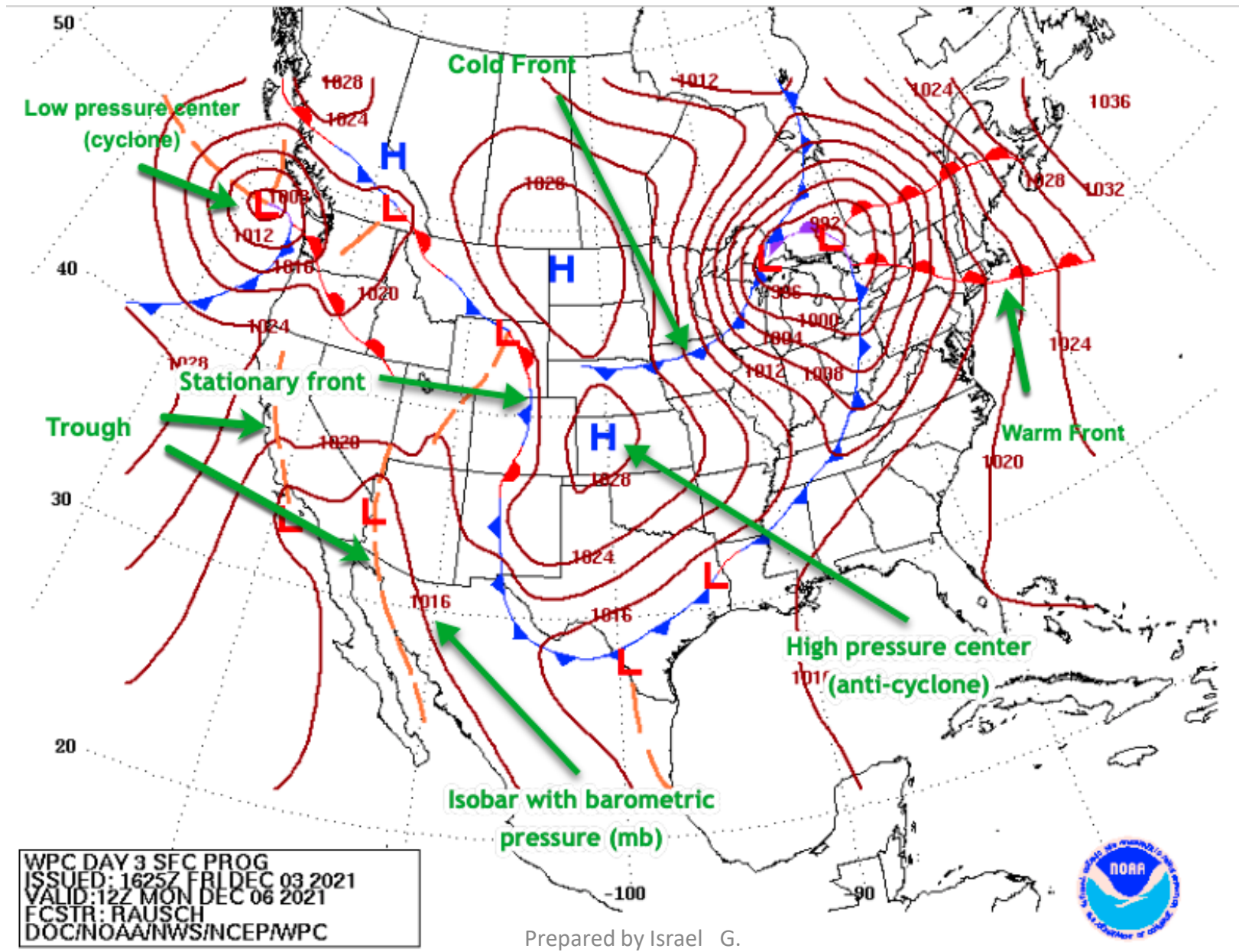


Weather Charts

Weather symbols and signs

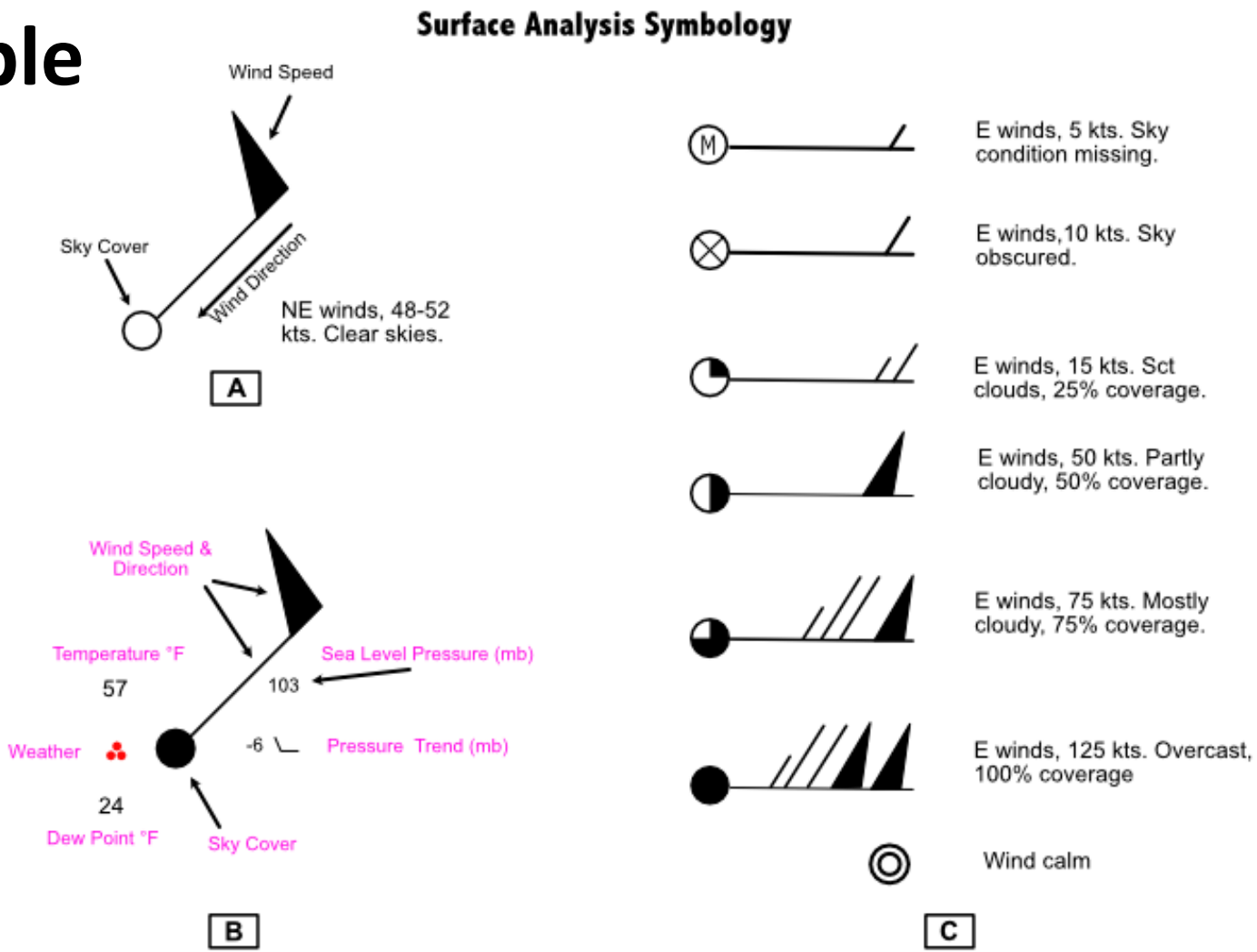


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Cont...

Example



Cont...

- As indicated in **A** above, the wind direction **blows toward the sky cover** symbol.
- Note also that **upper-level wind** charts display winds similarly to the wind symbol above, except without the **sky cover** symbol.
- A station plot, shown in **B** above,
 - ✓ displays temperature (57°F),
 - ✓ dew point (24°F),
 - ✓ the weather occurring at the time of observation (moderate rain),
 - ✓ sky cover (overcast),
 - ✓ wind (48-52 kts),
 - ✓ sea level pressure (1010.3 mb), and the pressure trend.
 - ✓ In this example, the pressure has dropped **six** millibars in the last **three hours** and the trend line is a **decrease** in pressure followed by **steady pressure**.

Cont...

- Barometric pressure is read in millibars, with **1013** mb representing **29.92"** of mercury, standard pressure on a standard **15°C** day.

- The leading **9** or **10** is omitted.

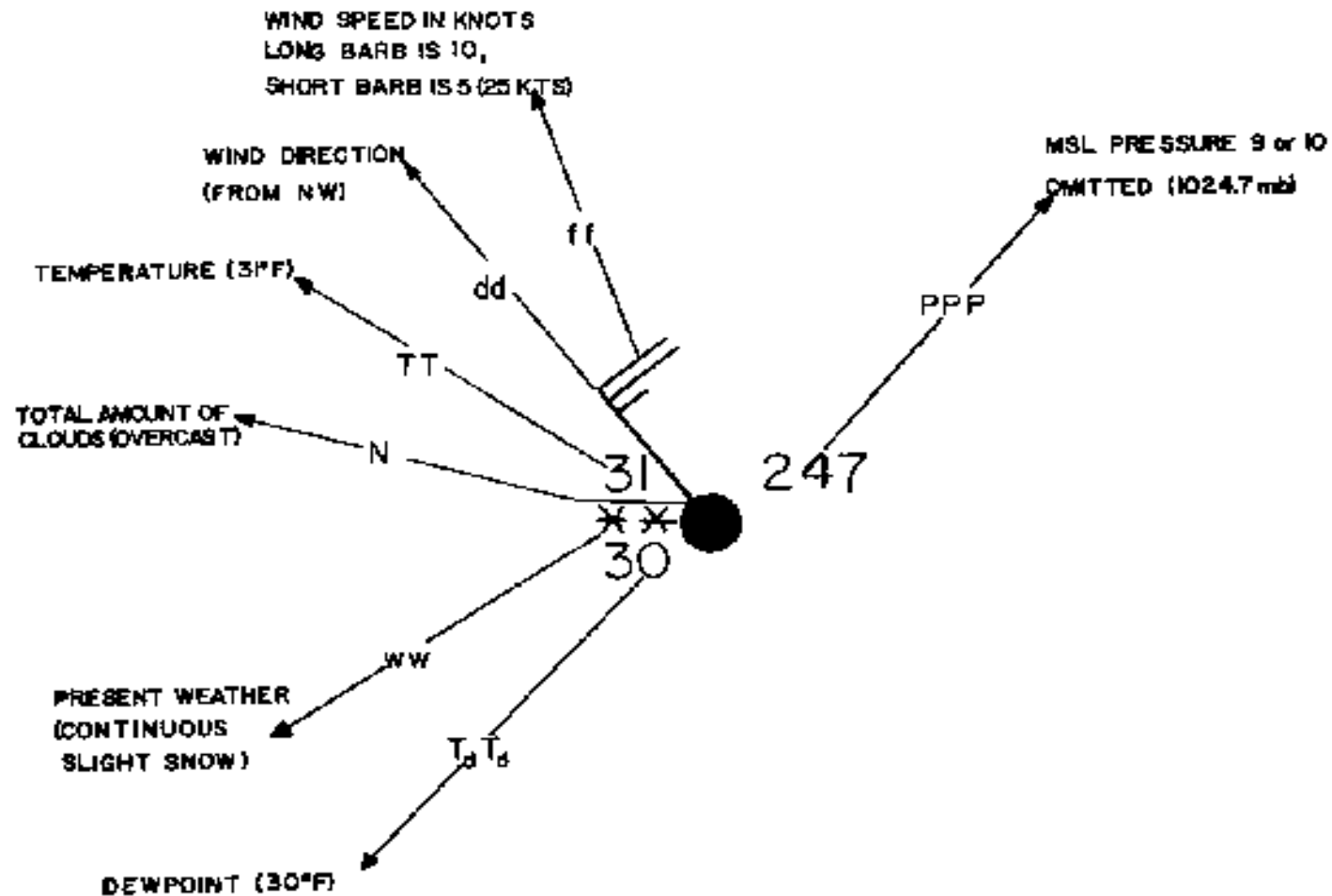
- Typical pressures are abbreviated on the charts as follows:

410—1041.0 mb 987—998.7 mb

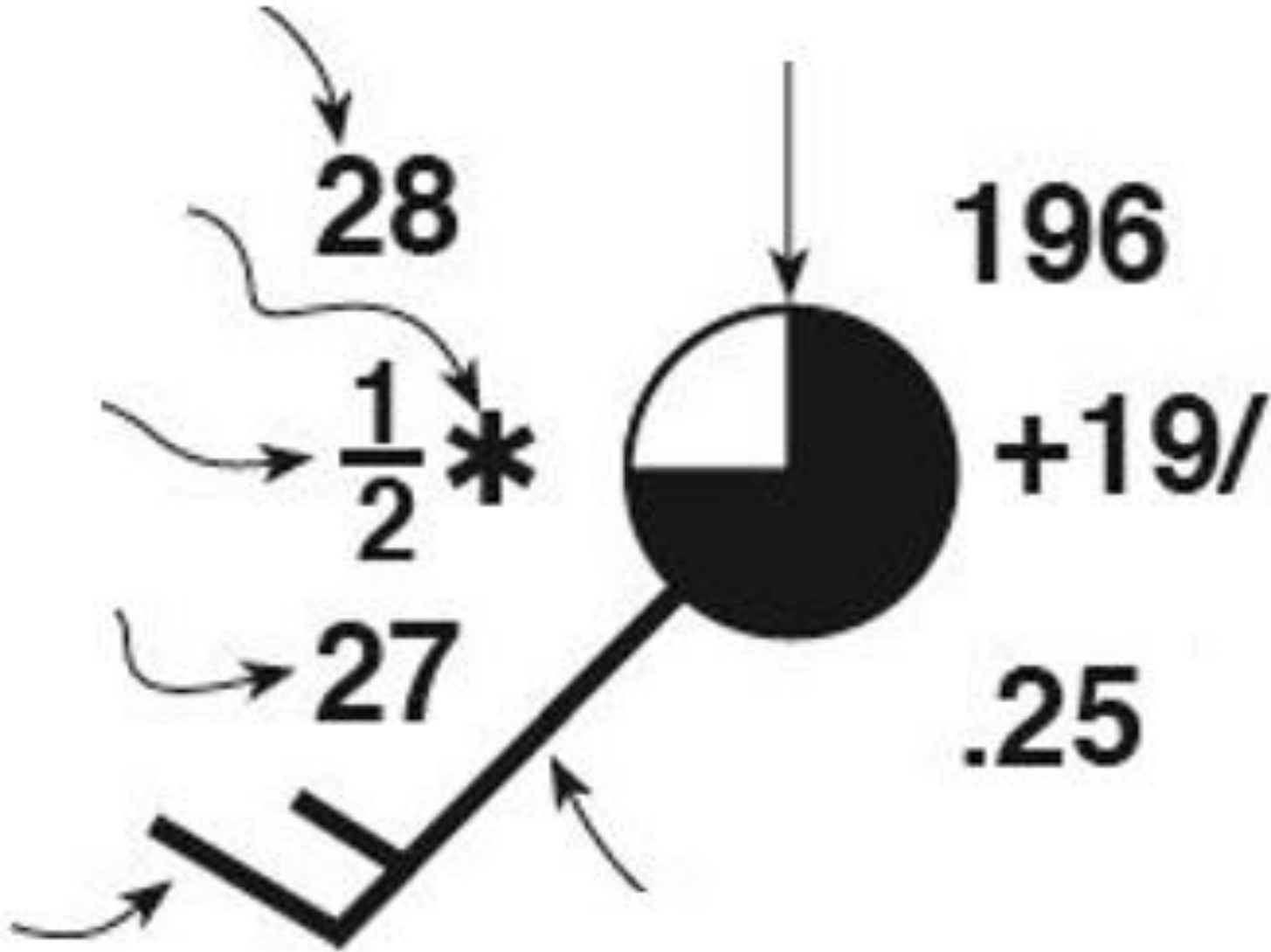
103—1010.3 mb 872—987.2 mb

- Note the convention. Millibars are always to **one decimal** place. A low pressure beginning with “9” or an “8” will have a “**9**” placed in front.
- Higher pressures will have a “**10**” added to the front.
- **C**, above, shows examples of **sky cover**, **wind speed** and **direction**, **obscuration**, and **missing sky** conditions.

Example

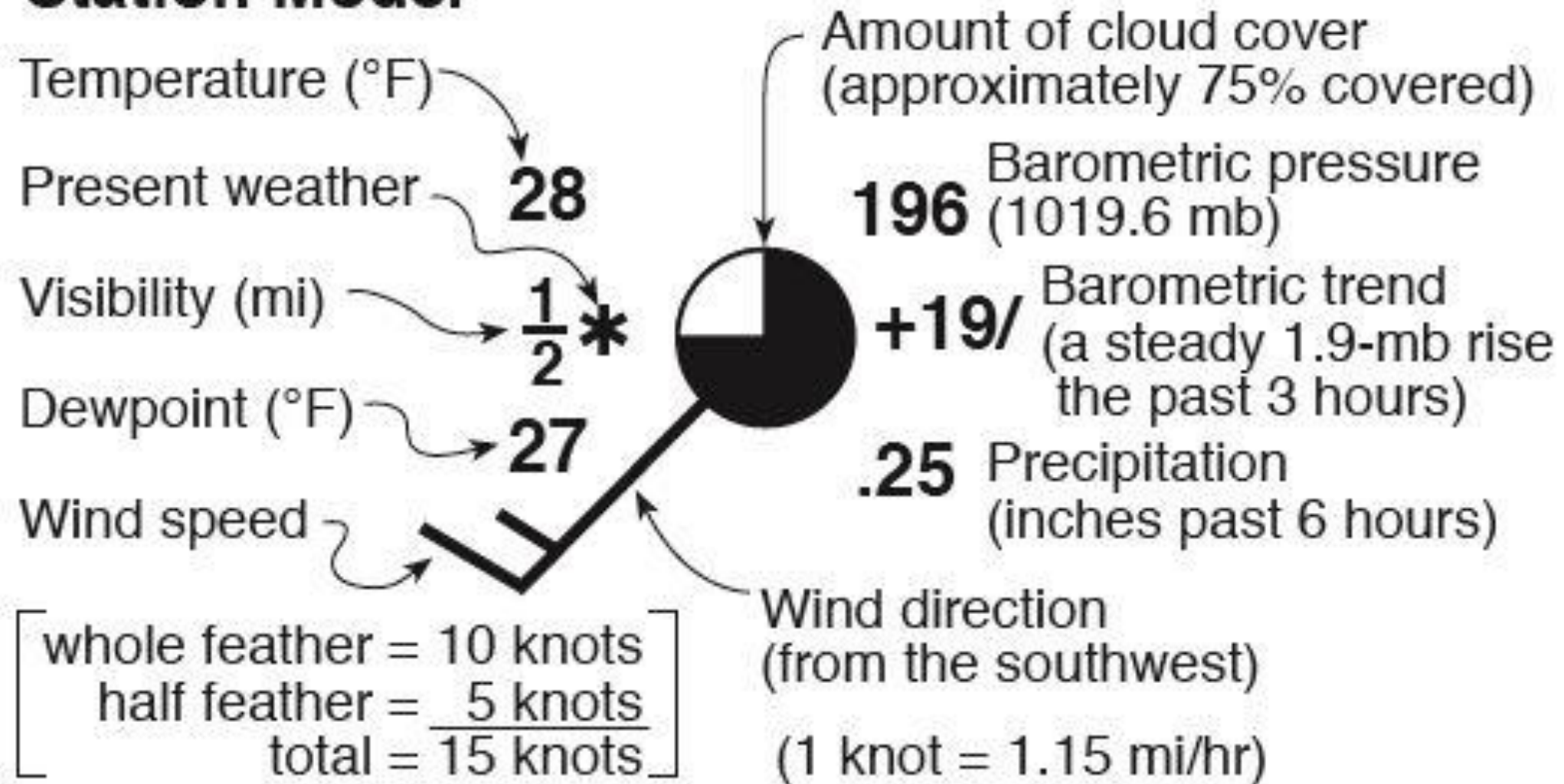


Exercise

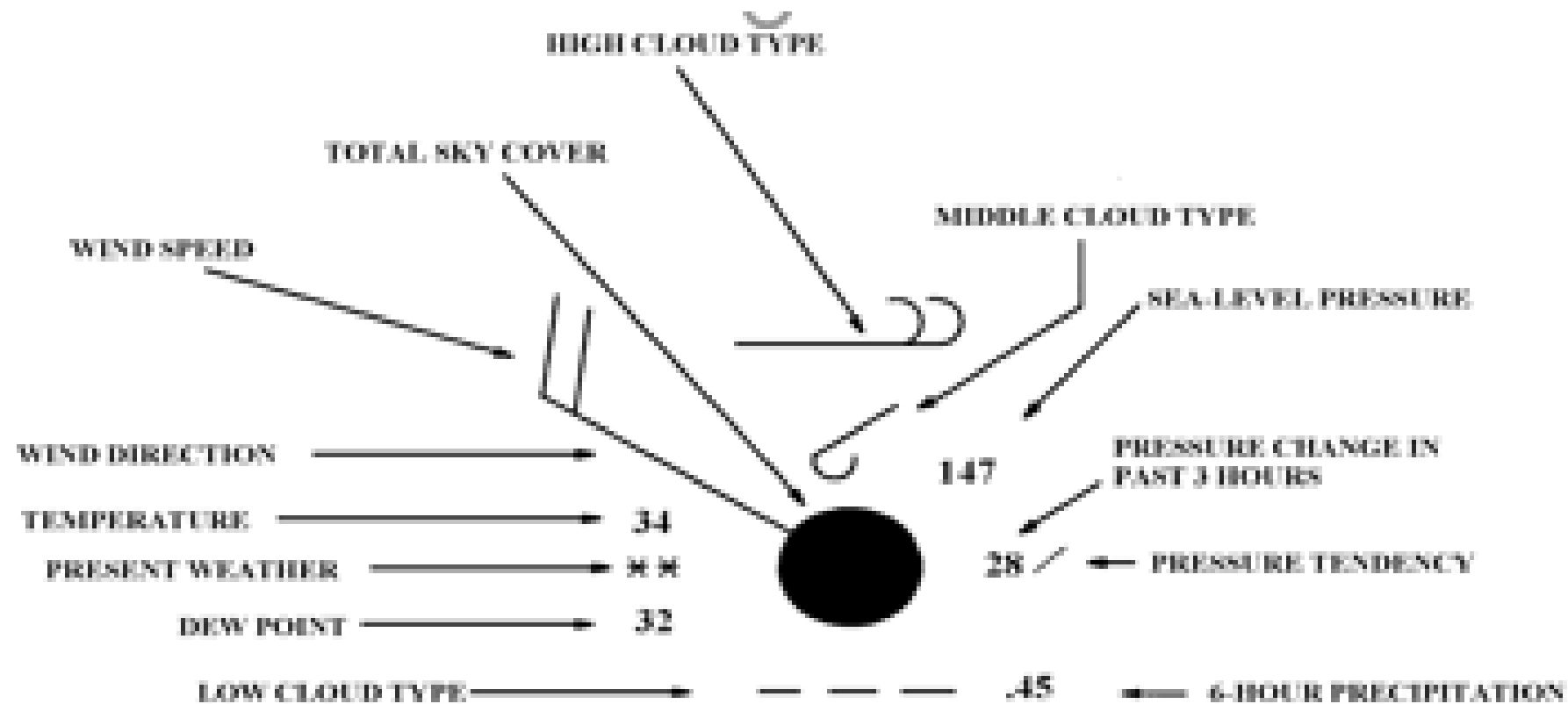


Answer

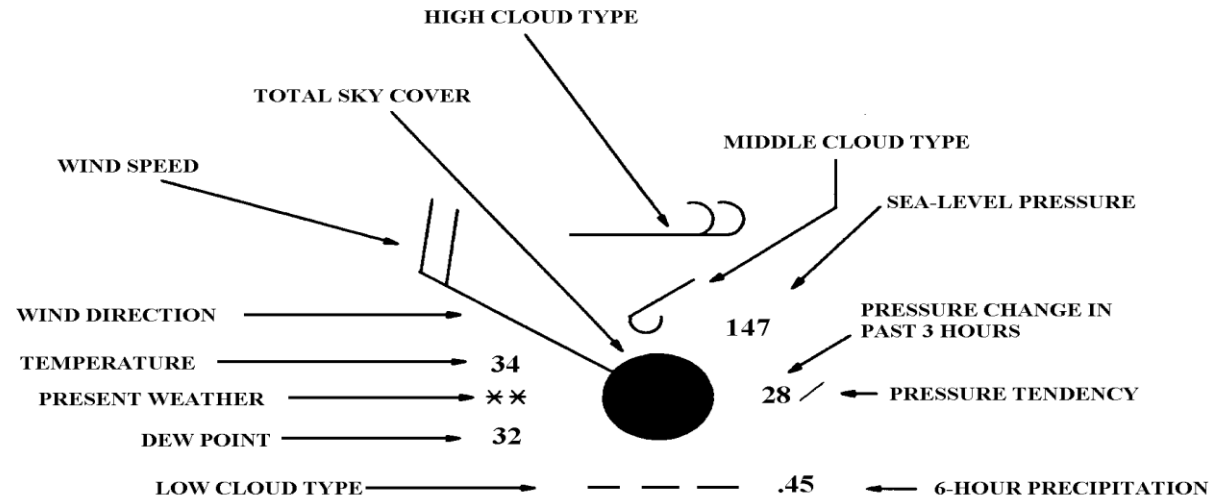
Station Model



Exercise

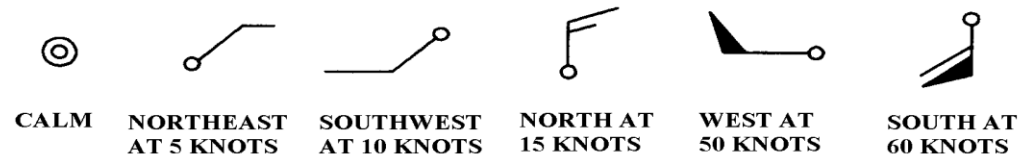


Answer



1. Total sky cover: Overcast.
2. Temperature: 34 degrees F, Dew Point: 32 degrees F.
3. Wind: From the northwest at 20 knots (relative to true north).

Examples of wind direction and speed



4. Present Weather: Continuous light snow.
5. Predominate low, middle, high cloud reported: Strato fractus or cumulus fractus of bad weather, altocumulus in patches, and dense cirrus.
6. Sea-level pressure: 1,014.7 millibars (mbs).
NOTE: Pressure is always shown in three digits to nearest tenth of an mb. For 1,000 mbs or greater, prefix a "10" to the three digits. For less than 1,000 mbs, prefix a "9" to the three digits.
7. Pressure change in the past 3 hours: Increased steadily or unsteadily by 2.8 mbs. The actual change is in tenths of a mb.
8. 6 - hour precipitation in hundredths of an inch: 45 hundredths of an inch.

Upper level air charts

- ✓ Upper-air weather maps differ from surface weather maps.
- ✓ **Whereas surface weather conditions are plotted on a map of constant altitude (normally sea- level), upper-air weather conditions are plotted on maps of constant air pressure.**
- ✓ The altitude at which the particular pressure is located is reported on these maps.

Cont...

- ✓ Surface and upper-air observations are looks identical at first inspection, there are huge fundamental differences between surface station plots and upper-level station plots.
- ✓ Station plots give information such as temperature, [dewpoint](#), cloud cover, weather, [pressure](#) (or height), [dewpoint depression](#), visibility, wind direction, wind speed, and height change.

Cont...

❑ The four biggest distinctions between surface and upper-air observations are:-

- ✓ the temperature units,
- ✓ the representation of dewpoint,
- ✓ the meaning of a filled in station plot circle, and
- ✓ the height/pressure representation.

Cont...

- ✓ Surface observations have the **dewpoint** and **temperature** plotted in degrees **Fahrenheit**.
- ✓ Observations plotted on an **upper-level chart** have the **temperature** and **dewpoint depression** plotted in units of **Celsius**.
- ✓ On a surface chart, **dewpoint** is plotted while on an upper-level chart the **dewpoint depression** is plotted.
- ✓ The **dewpoint depression** is the temperature **difference** between the **temperature** and the **dewpoint**.
- ✓ While the **dewpoint** is plotted directly on a surface plot it must be inferred on an upper air chart.
- ✓ The **dewpoint** is located below the **temperature** on a surface plot.
- ✓ The **dewpoint depression** is plotted below the **temperature** on an upper air plot.

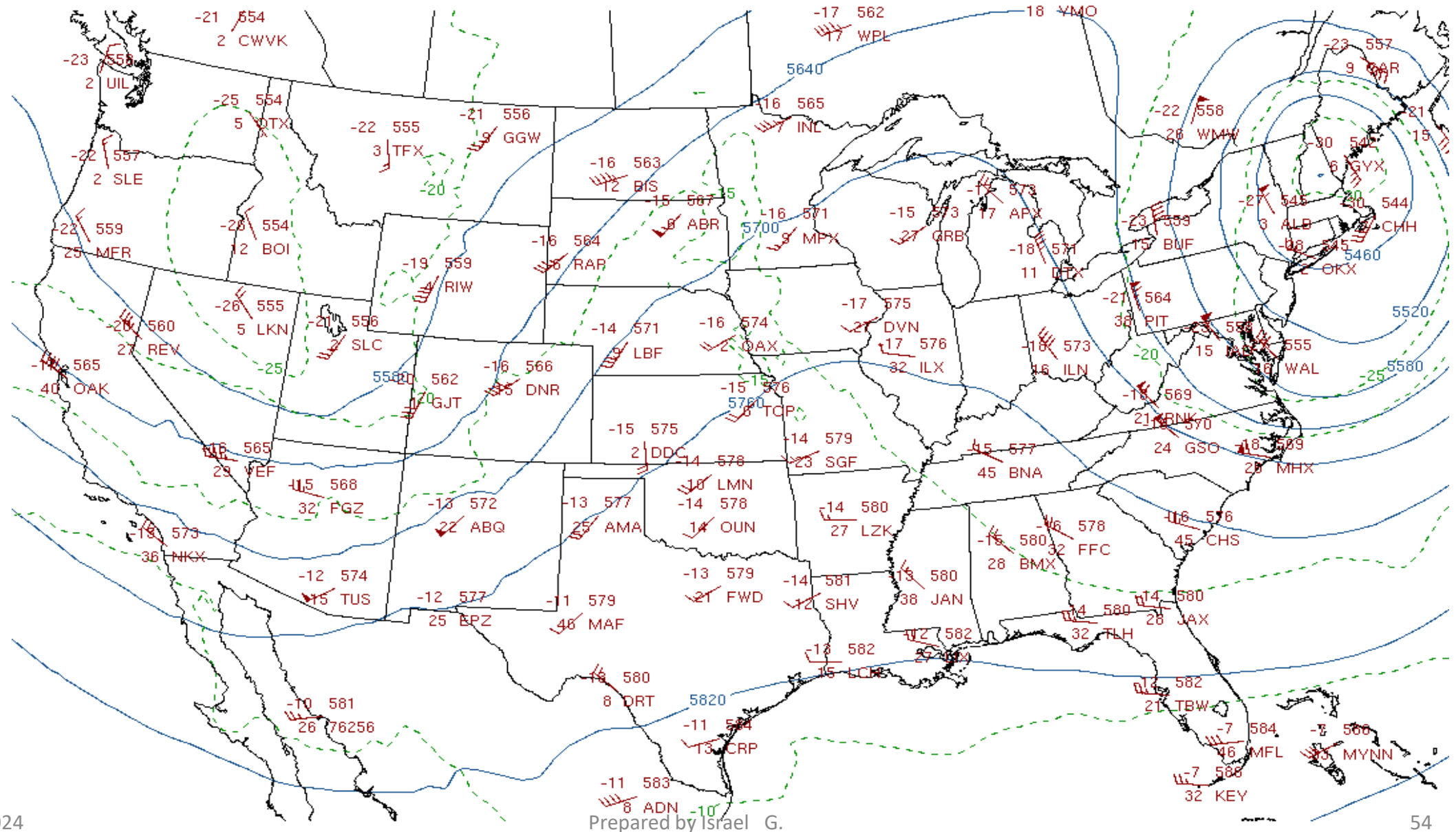
Cont...

- ✓ The meaning of a **filled in circle** is **cloudy** on a **surface** chart but means the **dewpoint depression** is **less than 6** on an **upper air chart**.
- ✓ On an **upper air chart** the **circle** will either be **blank** or **shaded** it.
- ✓ On a surface plot, the circle could be shaded none, 1/8, 2/8, 3/8, etc., or completely shaded (8/8) depending on how much **cloud cover** is present.
- ✓ A station plot shows **pressure** without the leading **9** or **10**. Example: **945** is **994.5** millibars while **034** is **1003.4** millibars.
- ✓ **Upper air plots** have a separate coding for each analysis **chart level**.

Cont...

- When the upper atmosphere is observed, there are specific pressure levels that are always reported.
- Above the surface, **barometric pressure** on weather charts is represented as constant pressure surfaces or the **height** (in meters) at which a particular **pressure value** is located.
- These levels are called mandatory pressure levels and are the surface, **850** mb, **700** mb, **500** mb, and **300** mb or **200** mb.
- Below is a look at a sample **500** mb chart.

Cont...



Cont...

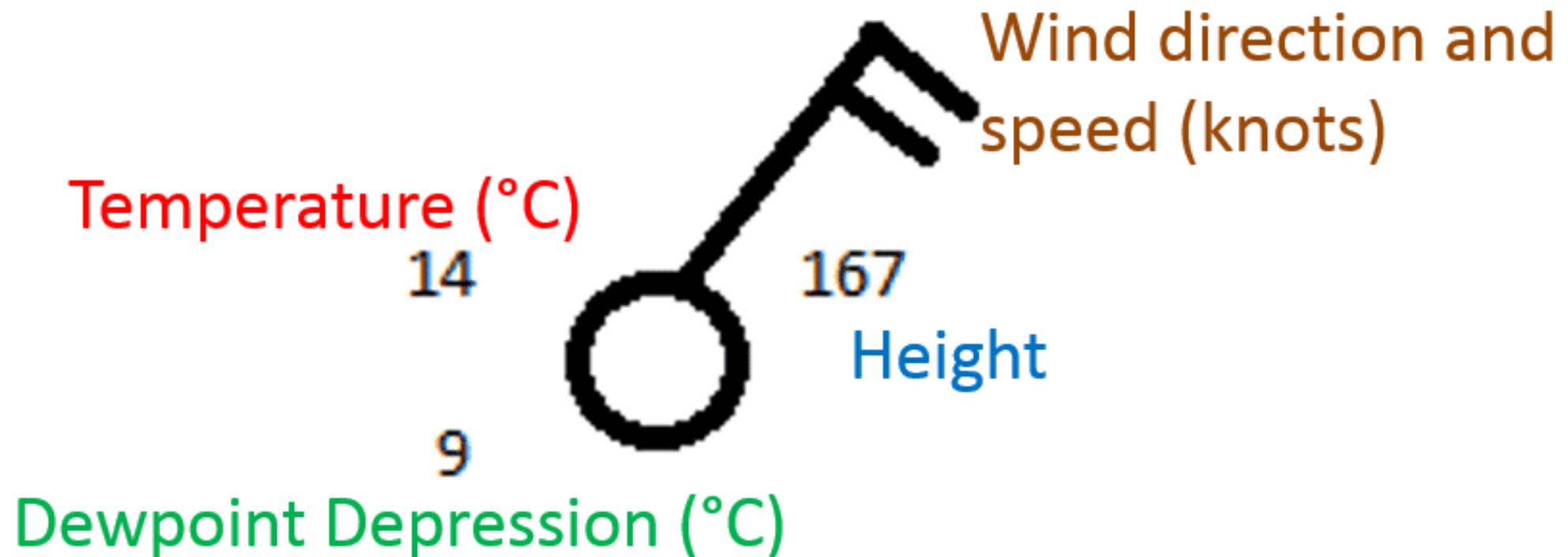
- The data plotted at a **point** on the map is called a **station plot** and is similar to one that you might see on a surface map.
- However, there are a few differences between the station plot for a surface and an upper air map.
- The height of the pressure surface is plotted in the top right of the station plot.

Cont...

- Depending on which **pressure level** you're looking at, there are **different** rules for **interpreting** the **height** of a station plot. These **rules** are summarized below.
- **850 mb**: Add 1000 to the value in the height field. Example: A value of 419 becomes a height of 1419 meters.
- **700 mb**: The leading number is either a **2** or **3**, whichever brings the value **closest** to **3000**. Example: **914** is **2,914** meters; **032** is **3,032** meters.
- **500 mb**: Add a **zero** to the end. Example: **548** becomes **5480** meters.
- **300 mb & 200 mb**: Add a **1** to the **front** and a **0** to **the end** Example: **065** is **10,650** meters.

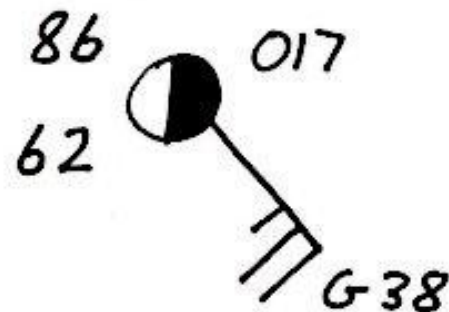
Cont...

- ❖ Below is an example of how a station plot may look on an upper air map at **850** mb pressure level.



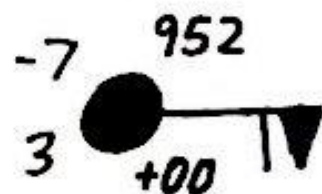
Example: Surface vs Upper air

#1 Surface Observation

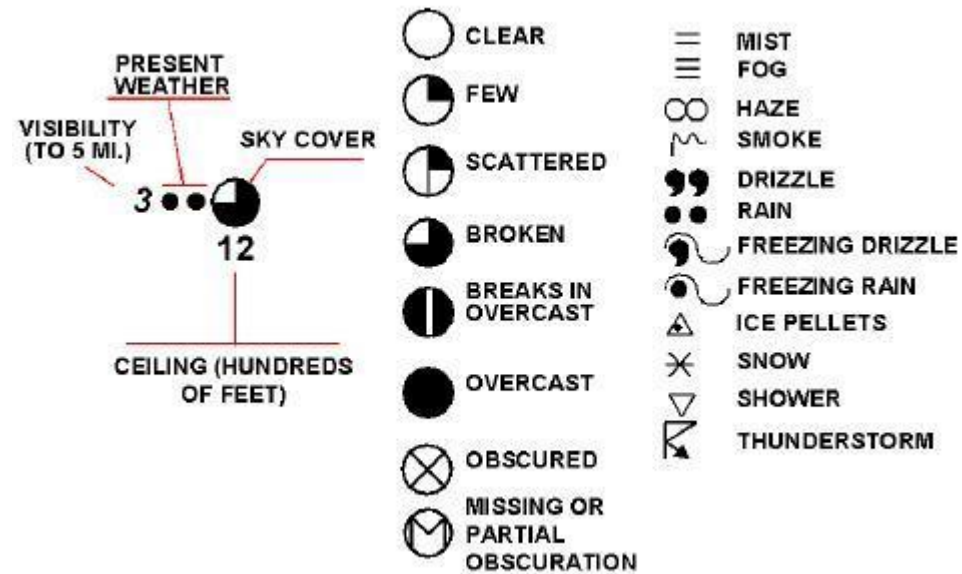


Temperature: 86°F Wind direction: *SE* Visibility: *unlimited*
Dewpoint: 62°F Wind speed: *25 knts* Dew point depression: *24*
Pressure: *1001.7 mb* Cloud cover: *1/2* Wind gust: *38 knts*

#2 Upper Air Observation



Temperature: -7°C Wind direction: *East*
Dewpoint: -10°C Wind speed: *60 knts* Dew point depression: *3*
Height: *2,952 gpm* Height change: *0 m* Shaded circle: *depression less than 5*



Symbol	Meaning	Symbol	Meaning
	Moderate turbulence		Rain shower
	Severe turbulence		Snow shower
	Moderate icing		Thunderstorms
	Severe icing		Freezing rain
	Rain		Tropical storm
	Snow		Hurricane (typhoon)
	Drizzle		

....END

Thanks !